

Young Bin Kim

M.S. STUDENT • CULTURE TECHNOLOGY, KAIST

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Research Interests

Augmented Reality, Mixed Reality, Spatial Computing, Human-Computer Interaction

Education

Korea Advanced Institute of Science and Technology (KAIST)

M.S. CULTURE TECHNOLOGY

Daejeon, South Korea

Mar. 2025 – Feb. 2027 (expected)

- Advisor: Prof. Woontack Woo, UVR Lab

Handong Global University

B.S. AI CONVERGENCE & ENTREPRENEURSHIP / B.A. INTERNATIONAL STUDIES

Pohang, South Korea

Mar. 2020 – Feb. 2025

- Cumulative GPA: 4.06 / 4.50, Major GPA: 4.18 / 4.50, *Cum Laude*
- Undergraduate research advisor: Prof. Daseong Han, Computer Graphics Lab

Research Experience

UVR Lab, KAIST

GRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. WOONTACK WOO

Daejeon, South Korea

Mar. 2025 – present

- On-the-fly integration of everyday objects into mixed reality; first-author manuscript in review. [R1]
- Built the user-study setup and analyzed the study results for a single-image 3D scene reconstruction system in mixed reality. [C1]
- Implemented the MR system and demo for a physics-aware scene-graph reconstruction system from a single view. [C2]
- Built an AR exhibition system that augments architectural elements from 2D artworks in the physical exhibition space. [P1]

Computer Graphics Lab, Handong Global University

UNDERGRADUATE RESEARCHER, ADVISOR: PROF. DASEONG HAN

Pohang, South Korea

2023 – 2024

- Built a video-based framework for whole-body and hand motion data collection and a finger-motion generation method for captured body motion. [D2, D3]

Publications

IN REVIEW

- [R1] **Young Bin Kim** et al. First-author manuscript in review at an international conference, 2027.

INTERNATIONAL CONFERENCES AND JOURNALS

- [C1] Seok-Young Kim, Dooyoung Kim, Taejun Son, **Young Bin Kim**, Woojin Cho, Woontack Woo. 2026. Mix3R: Zero-shot Interactive 3D Scene Reconstruction from a Single Image in Mixed Reality. *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, Proc. IEEE ISMAR 2026.
- [C2] Suji Kang, Seok-Young Kim, **Young Bin Kim**, Taewook Ha, Dieter Schmalstieg, Shohei Mori, Woontack Woo. 2026. SnapPhysics: A Physics-Aware Scene Graph from a Single View for Interactive Mixed Reality Scenes. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*.

INTERNATIONAL POSTERS

- [P1] **Young Bin Kim**, Suji Kang, Minju Baeck, Seonji Kim, Woontack Woo. 2025. Architectural Elements Augmentation from 2D Artworks for Spatial Experience of AR Exhibition. *IEEE ISMAR-Adjunct*, 861–862.

DOMESTIC CONFERENCES

- [D1] **Young Bin Kim**, Jinwoo Jeon, Woontack Woo. 2025. Instance-Aware Object-Centric 3D Map Generation Framework. *Korea Computer Congress (KCC)*, KIISE.
- [D2] Yunseop Kim, **Young Bin Kim**, Jina Lee, Jiseong Lee, Jaeho Lim, Daseong Han. 2024. A Video-based Framework for Collecting Whole-body and Hand Motion Data. *HCI Korea*, 87–91.

- [D3] Jiseong Lee, **Young Bin Kim**, Yunseop Kim, Jina Lee, Jaeho Lim, Daseong Han. 2024. Finger Motion Generation for Captured Whole Body Motion. *Korea Computer Graphics Society Conference (KCGS)*, 111–112. Poster.
- [D4] **Young Bin Kim**, Jiseong Lee, Gangsan Kim, Yunseop Kim, Jina Lee, Wonsup Lee. 2024. Measurement of VR Motion Sickness and Analysis of User Behavior Using VR Game and SSQ. *HCI Korea*, 37–42.

Research Projects

2025 – present	WISE UI: Wearable Interface for Sustainable Enhancement UVR Lab, KAIST Personalized UI/UX for smart glasses that overcomes the limits of current smart-glasses interfaces and supports sustainable enhancement of user abilities in daily life.
2025 – 2026	DT-XR: Development of Dynamic Digital Twin for Realistic Untact XR Collaboration UVR Lab, KAIST Building 3D spaces for dynamic digital twins that reconstruct static environments and reflect moving objects and cameras in real time.

Selected Projects

Spring 2026	ASL-based Shortcut Gestures for 3D Interaction Unity, hand tracking Built a 3D shortcut system mapping Ctrl+C/F/Z/X to ASL finger-spelling gestures, recognized in real time from both hands with left/right disambiguation and pinch-based block placement.
Fall 2025	K-Culture Stress Hunters Unity, AR; team lead Led a team to develop an AR system for implicit stress relief through three-stage embodied interaction based on Korean cultural motifs.
2024	Media Facade: “Under the Pohang Sea” Unreal Engine Developed underwater visuals and effects for a building-scale media facade; Grand Prize, Pohang Media Facade Hackathon.
2023	Legend of XR in the Metaverse of Madness Unity, VR; team lead Led the development of a three-quest VR experience for virtual-production training; built characters, VR interactions, and the intro cinematic. Excellence Award, COSS Metaverse Contest.
2023	EasyDizzy Unity, VR Designed and developed a first-person VR running game with a progressive-overload system for motion-sickness adaptation.

Awards & Honors

2024	Grand Prize (HGU President’s Award) , Pohang Culture & Arts Factory Hackathon, City of Pohang / Coding Ground / Handong Global University
2023	Creativity Award , Metaverse Asset Idea Contest, Korea Electronics Technology Institute
2023	Excellence Award , Metaverse Contest, COSS, Ministry of Education

Skills

Programming	C#, Python, Kotlin, C++
Software/Tools	Unity, Unreal Engine, OpenXR, PyTorch, OpenCV
Devices	Galaxy XR, Meta Quest 2/3, HoloLens 2
Languages	Korean (native), English (fluent)